DAY 12 – OSI MODEL

1. INTRODUCTION

The OSI (Open Systems Interconnection) Model is a conceptual framework developed by the International Organization for Standardization (ISO) to explain how data is transmitted between devices over a network.

The OSI model divides network communication into seven layers, with each layer performing specific functions.

Understanding the OSI model is important in cybersecurity because it helps identify network vulnerabilities, troubleshoot communication problems, and understand how security threats affect different layers.

2. OBJECTIVE

The main objective is to understand the seven layers of the OSI model and identify common protocols associated with each layer.

Specific Objectives

1. Understand the OSI model and its purpose.
2. Study the functions of all seven layers.
3. Identify protocols such as HTTP, TCP, IP, and Ethernet.
4. Understand how data travels through network layers.
5. Learn the importance of the OSI model in cybersecurity.

3. SEVEN LAYERS OF THE OSI MODEL

OSI Seven-Layer Architecture

7. Application

HTTP, HTTPS, DNS

6. Presentation

Encoding, encryption, compression

5. Session

Session establishment and management

4. Transport

TCP, UDP

3. Network

IP, ICMP

2. Data Link

Ethernet frames, MAC addressing

1. Physical

Cables, signals, physical media

Layer 7 is closest to user applications; Layer 1 handles physical transmission.

4. FUNCTIONS OF EACH OSI LAYER

Layer 7 – Application Layer

The Application Layer provides network services to user applications and allows software to communicate over a network.

Functions:

- Provides network services to applications.
- Supports web browsing and email communication.
- Enables access to network resources.

Examples: HTTP, HTTPS, DNS, SMTP, FTP.

Layer 6 – Presentation Layer

The Presentation Layer handles the representation and formatting of data so that communicating systems can interpret it.

Functions:

- Data translation
- Encoding and decoding
- Encryption and decryption
- Data compression

Examples: Character encoding such as UTF-8, data formats such as JPEG, and encryption-related functions. These are illustrative functions rather than protocols exclusive to Layer 6.

Layer 5 – Session Layer

The Session Layer manages communication sessions between applications.

Functions:

- Establishes communication sessions.
- Maintains sessions.
- Coordinates communication.
- Terminates sessions.

Examples: Session-management functions associated with RPC and NetBIOS Session Service.

Layer 4 – Transport Layer

The Transport Layer provides end-to-end communication between applications.

Functions:

- Data segmentation and reassembly
- Transport-level error handling
- Flow control where supported
- Port-based application identification

Protocols: TCP and UDP.

Layer 3 – Network Layer

The Network Layer is responsible for logical addressing and routing packets between networks.

Functions:

- IP addressing
- Packet forwarding
- Routing
- Network-level communication

Protocols: IPv4, IPv6, ICMP.

Devices: Routers and Layer 3 switches.

Layer 2 – Data Link Layer

The Data Link Layer handles communication over a local network link.

Functions:

- Frame formation
- MAC addressing
- Frame forwarding

- Error detection

Examples: Ethernet (IEEE 802.3), IEEE 802.11 MAC.

Devices: Ethernet switches and network interface cards.

Layer 1 – Physical Layer

The Physical Layer transmits raw bits through physical media using electrical, optical, or radio signals.

Functions:

- Bit transmission
- Signal representation
- Physical connections
- Transmission media specifications

Examples: Copper cables, fiber-optic connections, and physical signaling standards.

5. OSI LAYER COMPARISON CHART

Layer	Name	Main Function	Examples
7	Application	Application network services	HTTP, HTTPS, DNS
6	Presentation	Data formatting and representation	Encoding, encryption, compression
5	Session	Communication session management	RPC session functions
4	Transport	End-to-end transport	TCP, UDP
3	Network	Logical addressing and routing	IP, ICMP
2	Data Link	Framing and MAC addressing	Ethernet, 802.11 MAC
1	Physical	Transmission of raw bits	Copper, fiber, signals

6. DATA TRANSMISSION THROUGH THE OSI MODEL

When a user accesses a website, data passes through different network functions.

Sender Side:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Receiver Side:

Physical → Data Link → Network → Transport → Session → Presentation → Application

This process is called encapsulation and decapsulation.

Encapsulation

Encapsulation is the process of adding protocol information as data moves down the networking layers.

Decapsulation

Decapsulation is the process of removing and interpreting protocol information as data moves up the networking layers.

7. IMPORTANCE OF THE OSI MODEL IN CYBERSECURITY

The OSI model helps cybersecurity professionals understand threats and security controls at different networking layers.

Layer	Example Security Threat
Application	Web application attacks
Presentation	Weak encryption or unsafe data handling
Session	Session hijacking
Transport	TCP SYN flooding
Network	IP spoofing
Data Link	MAC spoofing and ARP poisoning
Physical	Cable tampering or physical interception

Understanding these threats helps professionals identify vulnerabilities and apply appropriate security controls.

8. OSI MODEL VS TCP/IP MODEL

OSI Model	TCP/IP Model
7 layers	Commonly represented using 4 layers
Conceptual reference model	Practical Internet protocol architecture
Separates application, presentation, and session	Combines these functions into the Application layer
Has separate data link and physical layers	Combines these into the Link layer

9. CONCLUSION

The OSI model provides a structured way to understand network communication through seven layers: Application, Presentation, Session, Transport, Network, Data Link, and Physical.

Each layer performs specific functions that contribute to successful data transmission. Common protocols such as HTTP, TCP, IP, and Ethernet can be mapped to their corresponding networking functions.

Understanding the OSI model is essential in cybersecurity because it helps identify network threats, analyze communication problems, and strengthen network security.